

## Arrival choices

Previous lesson reminder:

Start with 1 and 2. Try 3 and 4 when ready. Reading and exact scribing are available.

### 1 Are earthquakes caused by being in a warm climate zone?

Support: Use the reminder strip or talk it through with an adult.

My response

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### 2 Which meaning fits tectonic?

Support: Choose: Related to movement of Earth's plates OR People or things in harm's way.

My response

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### 3 What hazard followed seafloor displacement?

Support: Use the model evidence anchor A or read one relevant card together.

My response

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### 4 Does magnitude directly count people affected?

Support: Give a reason when ready.

My response

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Stretch: use a precise example or explain which detail supports your answer.

## Evidence cards

Read, point or listen. Keep the source label with any detail you use.

### A Japan 11 March 2011

A magnitude 9.1 earthquake occurred off north-east Honshu, Japan. Sudden movement on the subduction boundary displaced the seafloor and produced a tsunami.

Source: USGS event record, official20110311054624120\_30

Limit: Magnitude measures the earthquake; it is not a direct count of people affected.

### B Effects and responses

Shaking and coastal flooding damaged places and disrupted people's lives. Earthquake-resistant design addresses shaking; warnings, evacuation routes and higher ground address tsunami exposure.

Source: Authored explanation of the case and risk-management principles

Limit: A sea wall or warning cannot eliminate all risk. Do not judge people without considering access and warning time.

### C Heat and unequal risk

On 19 July 2022 the Met Office recorded 40.3°C at Coningsby, 40.2°C at St James's Park and 39.6°C at Topcliffe. Outdoor work increases exposure; difficulty accessing cool spaces can increase vulnerability. Shade and accessible cool spaces can reduce harm.

Source: Met Office July 2022 report, station table; risk explanation authored for teaching

Limit: This explains risk factors, not a medical assessment of an individual.

### D Reading a hazard feed

A USGS event row can include UTC time, place, magnitude and depth in kilometres. A short feed is a dated snapshot. It cannot predict the next earthquake.

Source: USGS earthquake feed documentation

Limit: Small samples, reporting thresholds and later revisions affect patterns.

## Visual resource

Use this visual with the evidence cards and enquiry questions.



Source: Natural Earth public-domain outlines, labels simplified. Terms of use:  
<https://www.naturalearthdata.com/about/terms-of-use/> (recorded in Sources and checks).

A detail I noticed and what it helps me explain

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# Data for the enquiry

Japan earthquake on 11 March 2011

| Event             | Magnitude |
|-------------------|-----------|
| Tohoku Japan 2011 | 9.1       |

Units: Magnitude

Source: USGS historical event record

Limit: Magnitude measures earthquake size. It does not count people affected or describe tsunami height.

My labelled graph or data notes

## Shared investigation

1. Locate the 2011 Japan event.
2. Build a cause, effect and response diagram using the case cards.

Work with a partner or adult, or use your own table. One person suggests; the other checks the evidence. Swap after one turn. Passing is allowed.

## Move or match the cards

| Cards to use                   |
|--------------------------------|
| Routes to higher ground        |
| Coastal flooding               |
| Sudden plate-boundary movement |

Possible labels: Cause / Effect / Response

Our explanation

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Support: Read one evidence card together. Highlight a useful detail. Rehearse with: The cause was \_\_\_\_.  
An effect was \_\_\_\_\_. A response could reduce \_\_\_\_ by \_\_\_\_\_.

Further challenge: Compare a response to shaking with a response to tsunami exposure.

## Supported independent task

Complete this route only. Describe how a natural hazard affects people and places.

1. Create a case-study diagram for Japan.
2. Interpret its map location and one data value.
3. Judge a response that addresses a stated effect or risk.

Support: Read one evidence card together. Highlight a useful detail. Rehearse with: The cause was \_\_\_\_\_. An effect was \_\_\_\_\_. A response could reduce \_\_\_\_\_ by \_\_\_\_\_. Complete the organiser one space at a time. The cause was \_\_\_\_\_. An effect was \_\_\_\_\_. A response could reduce \_\_\_\_\_ by \_\_\_\_\_.

Cause and supporting detail

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Event or change

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Consequence and supporting detail

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How the links connect

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An adult may read or scribe your exact response. They should not choose the answer.

## Standard independent task

Complete this route only. Describe how a natural hazard affects people and places.

1. Create a case-study diagram for Japan.
2. Interpret its map location and one data value.
3. Judge a response that addresses a stated effect or risk.

Support: The cause was \_\_\_\_\_. An effect was \_\_\_\_\_. A response could reduce \_\_\_\_\_ by \_\_\_\_\_.

Cause and supporting detail

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Event or change

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Consequence and supporting detail

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How the links connect

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Check the required evidence and revise one detail.

## Stretch independent task

Complete this route only. Describe how a natural hazard affects people and places.

1. Create a case-study diagram for Japan.
2. Interpret its map location and one data value.
3. Judge a response that addresses a stated effect or risk.
4. Compare a response to shaking with a response to tsunami exposure.

Support: Choose the evidence independently. Use the frame only if it helps.

Cause and supporting detail

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Event or change

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Consequence and supporting detail

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How the links connect

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Compare a response to shaking with a response to tsunami exposure.

## Exit ticket

Choose a response mode. You may give your answers privately to an adult.

### What hazard followed seafloor displacement?

My response

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### Does magnitude directly count people affected?

My response

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Support: The cause was \_\_\_\_\_. An effect was \_\_\_\_\_. A response could reduce \_\_\_\_\_ by \_\_\_\_\_.

Further challenge: Compare a response to shaking with a response to tsunami exposure.

One thing I changed or want to check

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